

Self-Dual Trace Confinement: A Six Birds Structural Law for Formed Closures Under Involutive Self-Duality

Ioannis Tsiokos 

Version 3: October 1, 2026

(Version 1: June 17, 2026)

Abstract

Let J be an involution of a measure space (X, μ) , and let ψ be a strongly measurable observable with values in a real Hilbert space Y that intertwines J with a linear isometric involution of Y . The odd part ψ_- of ψ vanishes on the fixed locus $\text{Fix}(J)$; it is called separating when it vanishes only there. When $\int_X \|\psi_-\|^2 d\mu < \infty$, the Gram operator $A_X = \int_X \psi_- \psi_-^* d\mu$ is positive and trace-class with $\text{tr } A_X = \int_X \|\psi_-\|^2 d\mu$. Consequently A_X is dominated in the Loewner order by positive trace-class operators B_n with $\text{tr } B_n \rightarrow 0$ if and only if $\psi_- = 0$ almost everywhere, and then, for a separating readout, μ -almost every point is fixed by J . The paper records this equivalence together with a Markov-type bound under quantitative separation, variants for domination on finite windows and for budgets with two defect terms, and Douglas's characterization of exact domination. The mathematics is elementary. Its purpose is to fix the precise form of a hypothesis in the author's Six Birds framework, Self-Dual Trace Confinement (SDTC): that in specified self-dual settings such domination records are supplied by independent structure, for instance by an operator built from other data. SDTC is a named hypothesis, not a theorem: for a given ledger with a separating odd readout it is equivalent to confinement, and it cannot hold for all involutive ledgers, as a complex-conjugation example shows. All theorems, lemmas and propositions are formalized in Lean 4 with Mathlib for real Hilbert spaces of arbitrary dimension, with the trace of a positive operator taken as the sum of its diagonal entries in a Hilbert basis and the Gram operator as a Bochner integral in the operator norm; the trace-class integrability of the integrand and the standard facts identifying this trace with the trace-class trace are used only in the text.

Keywords. involution; fixed locus; positive operators; trace-class operators; Loewner order; Douglas factorization; Self-Dual Trace Confinement; Six Birds Theory; Lean formalization.

1 Introduction

When must an obstruction attached to a symmetry vanish? Here is the form of the question studied in this paper. A space of objects X carries an involution J , so that applying J twice returns the original object, and a measure μ that records which objects are present. The objects are observed through a strongly measurable map ψ with values in a real Hilbert space Y , and Y carries a linear isometric involution J_{iso} compatible with ψ in the sense that $\psi(Jx) = J_{\text{iso}}\psi(x)$. The involution J_{iso} splits every response into an even part and an odd part. The odd part is given by the orthogonal projection

$$P_- = \frac{1}{2} (I_Y - J_{\text{iso}}), \quad \psi_-(x) = P_- \psi(x),$$

and it changes sign under the duality: $\psi_-(Jx) = -\psi_-(x)$. In particular ψ_- vanishes at every fixed point of J . When it vanishes *only* there, the readout ψ_- detects exactly the displacement of an object from the fixed locus $\text{Fix}(J)$.

The obstruction is the positive operator

$$A_X = \int_X \psi_-(x) \psi_-(x)^* d\mu(x),$$

where v^* denotes the functional $h \mapsto \langle v, h \rangle$, so that vv^* is the rank-one operator $h \mapsto \langle v, h \rangle v$. When $\int_X \|\psi_-\|^2 d\mu < \infty$, the operator A_X is positive and trace-class, and

$$\text{tr } A_X = \int_X \|\psi_-(x)\|^2 d\mu(x).$$

Suppose now that A_X is dominated by positive trace-class operators whose traces tend to zero:

$$A_X \preceq B_n \quad (n \geq 1), \quad \text{tr } B_n \rightarrow 0,$$

where $\preceq$ is the Loewner order ($A \preceq B$ means that $B - A$ is positive). Monotonicity of the trace gives $0 \leq \text{tr } A_X \leq \text{tr } B_n$ for every n , so $\text{tr } A_X = 0$. Hence $A_X = 0$ and $\psi_- = 0$ almost everywhere. If ψ_- vanishes only on $\text{Fix}(J)$, then μ -almost every object is fixed by J : the visible mass is *confined* to the fixed locus. Conversely, if $\psi_- = 0$ almost everywhere then $A_X = 0$ and $B_n = 0$ will do. The equivalence is immediate; it is recorded as Theorem 7.1.

The argument is short, and none of its operator-theoretic ingredients is new: the Loewner order [7, 1], trace ideals [9], Markov's inequality [4], the arithmetic–geometric mean inequality [5] and Douglas's factorization theorem [3] are classical. Nor does the theorem make confinement easier to prove: by the equivalence just stated, producing domination records is exactly as hard as proving $\int_X \|\psi_-\|^2 d\mu = 0$. What the formulation offers is a format for the case in which A_X is not accessible directly. In applications of the framework one computes, from other data, a positive operator K^- on Y —for instance a carrier-side quantity built, in finite dimensions, from an audit operator and a probe [15]—and establishes, by a separate argument, a bridge inequality $A_X \preceq K^- + E$ with an explicit positive defect E . Trace bounds on K^- and E then control $\text{tr } A_X = \int_X \|\psi_-\|^2 d\mu$, the odd energy of the ledger; under quantitative separation this in turn bounds the mass at a given distance from the fixed locus. All of the substance lies in the bridge; the results of this paper record what follows from it.

The paper therefore separates two things that are easy to conflate:

1. the *confinement theorem* and its variants, proved here, which convert domination records into confinement on the fixed locus; and
2. a *hypothesis* about which structures possess such records. In the author's Six Birds framework [12] this hypothesis is named Self-Dual Trace Confinement, written Γ_{SDTC} (SDTC for short): in specified closures that carry a genuine involutive self-duality, trace-vanishing domination records for the odd readout are supplied by the structure. SDTC is not proved here, and no result of this paper depends on it. It is not a statement about all involutive ledgers: the first example in Section 3, taken with Lebesgue measure on a disc centred on the real axis, is an involutive ledger with a separating odd readout that is not confined, and hence has no such records.

Section 2 explains the framework vocabulary in plain terms; a reader interested only in the operator theory can skip it.

A guiding picture is the functional-equation symmetry of the Riemann zeta function. The map $s \mapsto 1 - \bar{s}$ is an involution of the complex plane whose fixed locus is the critical line $\text{Re } s = \frac{1}{2}$,

and $\operatorname{Re} s - \frac{1}{2}$ is the coordinate that changes sign. A measure on the nontrivial zeros, weighted by multiplicity, is confined to the fixed locus exactly when the Riemann hypothesis holds. The picture explains the shape of the abstraction; no result about ζ is used or claimed in this paper. A companion paper [16] studies this instance under explicitly stated hypotheses.

Results. Throughout, $(X, J, \mu, \psi, Y, J_{\text{iso}})$ is an involutive object ledger (Definition 3.1) with $\int_X \|\psi_-\|^2 d\mu < \infty$.

1. *Trace identity* (Lemma 4.3). A_X is a positive trace-class operator with $\operatorname{tr} A_X = \int_X \|\psi_-\|^2 d\mu$.
2. *Direct confinement* (Theorems 5.1 and 5.2). If $A_X = 0$ and ψ_- is separating, μ -almost every point is fixed by J . If X is a metric space and $\|\psi_-(x)\| \geq m(\varepsilon) > 0$ whenever $\operatorname{dist}(x, \operatorname{Fix}(J)) \geq \varepsilon$, then the set of points at distance at least ε from $\operatorname{Fix}(J)$ has outer measure at most $\operatorname{tr} A_X / m(\varepsilon)^2$.
3. *Douglas factorization* (Theorem 6.2). For bounded operators V, W with a common domain, $V^*V \preceq W^*W$ if and only if $V = TW$ for a contraction T . Applied to the analysis operator of ψ_- , this describes exact domination of A_X as a factorization problem.
4. *Confinement theorem* (Theorem 7.1). If $A_X \preceq B_n$ for positive trace-class B_n with $\operatorname{tr} B_n \rightarrow 0$, then $A_X = 0$, $\psi_- = 0$ almost everywhere, and, if ψ_- is separating, μ -almost every point is fixed by J . Conversely, if $\psi_- = 0$ almost everywhere, the zero sequence is such a family of records.
5. *Finite windows and defect budgets* (Theorem 8.2 and Proposition 8.4). The same conclusion holds when the domination is supplied on finite windows: real Hilbert spaces Y_n , bounded operators $\iota_n : Y_n \rightarrow Y$, positive trace-class operators $A_{X,n} \preceq B_n$ on Y_n , and positive trace-class operators T_n on Y , with $A_X \preceq \iota_n A_{X,n} \iota_n^* + T_n$ and $\operatorname{tr}(\iota_n B_n \iota_n^*) + \operatorname{tr} T_n \rightarrow 0$. When A_X is dominated by $(1+t)(K + E_{\text{src}}) + (1+t^{-1})E_{\text{br}}$ for every $t > 0$, with $K, E_{\text{src}}, E_{\text{br}}$ positive trace-class, then $\operatorname{tr} A_X \leq (\sqrt{a} + \sqrt{b})^2$ with $a = \operatorname{tr}(K + E_{\text{src}})$, $b = \operatorname{tr} E_{\text{br}}$, and this value is the infimum over $t > 0$ of the scalar budgets.

Every numbered result in this list, and the abstract squeeze of Remark 7.2, is formalized in Lean 4 with Mathlib for real Hilbert spaces of arbitrary dimension. The formal proofs take the trace of a positive operator to be the sum of its diagonal entries in a Hilbert basis and the Gram operator to be a Bochner integral in the operator norm. Bochner integrability in the trace class, the standard facts identifying the two traces, and the application of Douglas factorization to the analysis operator are checked only in the text. Appendix A lists the correspondence.

Organization. Section 2 explains the framework vocabulary. Sections 3 and 4 define the ledger, the odd readout, the operator A_X and separation. Section 5 proves the direct confinement results, Section 6 discusses domination and Douglas factorization, Section 7 proves the confinement theorem, and Section 8 gives the finite-window and defect-budget refinements. Sections 9 and 10 discuss scope and possible instances, and Section 11 concludes.

2 Framework

This section explains the few framework terms used in the paper. None of them enters a definition, statement or proof in Sections 3 to 8; a reader interested only in the operator theory may skip to Section 3.

Closures. The author’s Six Birds framework [12] describes a mathematical theory as the collection of fixed points of an idempotent operator: a completion rule that, applied twice, gives the same result as applied once. The fixed points are the objects that the theory regards as complete, or *closed*. Later papers add a discipline for which candidate objects are admissible [13] and a finite vocabulary for what an observer can see and certify about them [14]. The title of this paper speaks of *formed closures*; in the body the phrase is used only informally, for a concrete structure—here an involutive object ledger—that an application has specified completely. No definition or result below depends on it.

Structural laws and recognition sources. The framework distinguishes two kinds of statement. A *theorem* is proved from stated hypotheses in the usual way. A *structural law* is a named hypothesis asserting that the structures of a specified class carry certain data. It is stated separately, with a name, so that every result that uses it can be seen to use it. The data whose existence the law asserts are called its *recognition source*: the part of an argument that the structure itself must supply and that is not derived from anything else.

In this paper the structural law is Self-Dual Trace Confinement, whose name is abbreviated Γ_{SDTC} or SDTC:

In a specified class of closures that carry a genuine involutive self-duality, observed through a separating odd readout, the structure supplies positive trace-class operators B_n with $A_X \preceq B_n$ and $\text{tr } B_n \rightarrow 0$.

The recognition source is the sequence (B_n) . The word “genuine” is not given a formal meaning, and the class of closures must be specified in each application; the hypothesis has content only once both are fixed. It cannot be asserted for all involutive ledgers: the complex-conjugation example of Section 3 with Lebesgue measure on a disc centred on the real axis is an involutive ledger with a separating odd readout whose mass is not confined, so by Theorem 7.1 it has no such sequence. For any single ledger with a separating odd readout, SDTC is equivalent to confinement (Section 9). Nothing in this paper establishes the hypothesis in any instance.

What the argument uses. The confinement theorem uses the operator A_X only through its trace, the Loewner order and the fact that a positive operator of trace zero is zero. This is why one statement covers finite weighted configurations, measure spaces and, in Remark 7.2, abstract ordered sets with a faithful trace.

3 Involutive object ledger

The data of the paper are a measured space of objects with an involution, and an observable compatible with it.

Definition 3.1 (Involutive object ledger). An *involutive object ledger* is a tuple

$$\hat{\mathfrak{X}} = (X, J, \mu, \psi, Y, J_{\text{iso}})$$

consisting of

1. a measurable space X , the *carrier*;
2. an involution $J : X \rightarrow X$, that is, $J(Jx) = x$ for every $x \in X$;
3. a positive measure μ on X ;

4. a real Hilbert space Y , the *response space*;
5. a linear isometric involution $J_{\text{iso}} : Y \rightarrow Y$, that is, $J_{\text{iso}}^2 = I_Y$ and $\langle J_{\text{iso}}v, J_{\text{iso}}w \rangle = \langle v, w \rangle$ for all $v, w \in Y$;
6. a strongly measurable map $\psi : X \rightarrow Y$, the *readout*, that is *equivariant*:

$$\psi(Jx) = J_{\text{iso}}\psi(x) \quad (x \in X).$$

The *fixed locus* of the ledger is $\text{Fix}(J) = \{x \in X : Jx = x\}$.

The carrier-side involution J acts on the objects whose mass is being studied, while J_{iso} acts on the space in which those objects are observed; equivariance says that observing Jx is the same as observing x and then applying J_{iso} . Strong measurability (being a pointwise limit of measurable simple functions) is the standard condition under which Bochner integrals of Y -valued maps are defined; when Y is separable it is equivalent to Borel measurability. Every result below remains true if ψ is only μ -almost everywhere strongly measurable (measurability statements then hold up to μ -null sets), and the formal verification uses that weaker hypothesis. The measure μ is not assumed σ -finite; under the standing hypothesis $\int_X \|\psi_-\|^2 d\mu < \infty$ of the later sections, the set $\{\psi_- \neq 0\}$ is automatically σ -finite. No measurability of J and no invariance of μ under J is assumed: none of the results below needs them.

The finite weighted case. If X is a finite set with all subsets measurable and every point has finite mass, the measure is a family of real weights $\mu(x) \geq 0$, and integrals are finite sums; this is the *finite weighted case*. The *visible support* of μ is the set of objects of positive weight. In general, a statement holds for μ -almost every x when the set where it fails is contained in a measurable set of measure zero; in the finite case this means that it holds at every visible object. The mass of the ledger is *confined to the fixed locus* when $Jx = x$ for μ -almost every x , that is, when $X \setminus \text{Fix}(J)$ is μ -null. If $\text{Fix}(J)$ is measurable this says $\mu(X \setminus \text{Fix}(J)) = 0$.

Example (Complex conjugation). Let $X \subset \mathbb{C}$ be a Borel set stable under complex conjugation, with μ any positive Borel measure on X . Put

$$J(z) = \bar{z}, \quad Y = \mathbb{R}^2, \quad J_{\text{iso}}(u, v) = (u, -v), \quad \psi(z) = (\text{Re } z, \text{Im } z).$$

Then $\psi(Jz) = (\text{Re } z, -\text{Im } z) = J_{\text{iso}}\psi(z)$, so the readout is equivariant. The fixed locus is $X \cap \mathbb{R}$, and the coordinate that changes sign is the imaginary part. Mass is confined to the fixed locus exactly when μ is carried by the real axis.

Example (Zeros of the Riemann zeta function). The same pattern underlies the motivating example. The map $J_L(s) = 1 - \bar{s}$ is an involution of $\mathbb{C}$ with fixed locus the critical line $\text{Re } s = \frac{1}{2}$. Taking for X the nontrivial zeros of the Riemann zeta function, μ the counting measure weighted by multiplicity (the zeros are symmetric under J_L by the functional equation and the reflection principle [11]), $Y = \mathbb{R}$, $J_{\text{iso}} = -I$ and $\psi(s) = \text{Re } s - \frac{1}{2}$, one obtains an involutive object ledger whose mass is confined to the fixed locus exactly when the Riemann hypothesis holds. Here ψ is already odd, so $\psi_- = \psi$. This example only illustrates the definition. The results below also require $\int_X \|\psi_-\|^2 d\mu = \sum_\rho (\text{Re } \rho - \frac{1}{2})^2 < \infty$, summed over the zeros with multiplicity; the sum vanishes if the Riemann hypothesis holds, but its finiteness is not known otherwise. Nothing in this paper bears on the location of the zeros.

4 Anti-invariant ledger and separation

Fix an involutive object ledger $\hat{\mathfrak{X}} = (X, J, \mu, \psi, Y, J_{\text{iso}})$. This section extracts the odd part of the readout, builds the positive operator A_X from it, and computes its trace.

The odd readout. Because J_{iso} is a linear isometric involution it is self-adjoint: $J_{\text{iso}}^* = J_{\text{iso}}^{-1} = J_{\text{iso}}$. Hence

$$P_- = \frac{1}{2} (I_Y - J_{\text{iso}})$$

satisfies $P_-^2 = P_- = P_-^*$; it is the orthogonal projection onto the (-1) -eigenspace $\{v \in Y : J_{\text{iso}}v = -v\}$. The *anti-invariant* (or odd) readout is

$$\psi_-(x) = P_- \psi(x) \quad (x \in X).$$

It is strongly measurable, and by equivariance

$$\psi_-(Jx) = P_- J_{\text{iso}} \psi(x) = -P_- \psi(x) = -\psi_-(x),$$

because $P_- J_{\text{iso}} = -P_-$. In particular, if $Jx = x$ then $\psi_-(x) = -\psi_-(x)$, so $\psi_-(x) = 0$: the odd readout vanishes on the fixed locus. The converse is the separation property defined below. Only this vanishing is used later: none of the results needs J to be an involution, and all of them hold for any map $J : X \rightarrow X$ with $\psi \circ J = J_{\text{iso}} \circ \psi$. The involutive setting is the one in which the fixed locus has the meaning of a self-dual locus, and we keep it for that reason.

Positive and trace-class operators. On a real Hilbert space an operator T is *positive*, written $T \succeq 0$, if it is self-adjoint and $\langle Th, h \rangle \geq 0$ for all h ; the Loewner order is $A \preceq B$ if and only if $B - A \succeq 0$. For $v \in Y$ write vv^* for the rank-one operator $h \mapsto \langle v, h \rangle v$; it is positive. Let $\mathcal{S}_1(Y)$ denote the trace class with its norm $\|\cdot\|_1$. We use four standard facts [9, 8]:

- (T1) $\mathcal{S}_1(Y)$ is a Banach space, and the trace is a continuous linear functional with $|\text{tr } T| \leq \|T\|_1$; for $v \in Y$, $\|vv^*\|_1 = \text{tr}(vv^*) = \|v\|^2$.
- (T2) For $h, k \in Y$, the map $T \mapsto \langle Th, k \rangle$ is a continuous linear functional on $\mathcal{S}_1(Y)$.
- (T3) If $T \succeq 0$ is trace-class, then $\text{tr } T = \sum_i \langle Te_i, e_i \rangle \geq 0$ for every orthonormal basis (e_i) , and $\text{tr } T = 0$ implies $T = 0$.
- (T4) If $0 \preceq A \preceq B$ and B is trace-class, then A is trace-class and $\text{tr } A \leq \text{tr } B$.

These are usually stated for complex Hilbert spaces; for a real Hilbert space they follow by passing to the complexification $Y \otimes_{\mathbb{R}} \mathbb{C}$, which preserves positivity of self-adjoint operators, the trace norm and the trace. When Y is finite-dimensional every operator is trace-class and (T1)–(T4) are facts of linear algebra. The formal verification (Appendix A) does not use (T1)–(T4): it proves the results with the trace of a positive operator defined as $\sum_i \langle Te_i, e_i \rangle \in [0, \infty]$. By (T3) this agrees with the trace on positive trace-class operators, and a positive operator for which it is finite is trace-class [8].

Definition 4.1 (Anti-invariant object ledger). Assume $\int_X \|\psi_-(x)\|^2 d\mu(x) < \infty$. The *anti-invariant object ledger* of $\hat{\mathfrak{X}}$ is the operator

$$A_X = \int_X \psi_-(x) \psi_-(x)^* d\mu(x),$$

the integral being a Bochner integral in $\mathcal{S}_1(Y)$.

The integral exists by Lemma 4.3 below. For a finite weighted ledger it is the finite sum

$$A_X = \sum_{x \in X} \mu(x) \psi_-(x) \psi_-(x)^*,$$

a nonnegative combination of rank-one positive operators, one for each visible object, each recording the part of the readout of that object that changes sign under the duality.

Definition 4.2 (Separating anti-invariant readout). The odd readout ψ_- is *qualitatively separating* if, for μ -almost every $x \in X$,

$$\psi_-(x) = 0 \iff x \in \text{Fix}(J).$$

In the finite weighted case this means that the equivalence holds at every object of positive weight.

Suppose in addition that X is a metric space. The odd readout is *quantitatively separating* with modulus $m : (0, \infty) \rightarrow (0, \infty)$ if, for every $\varepsilon > 0$ and every $x \in X$,

$$\text{dist}(x, \text{Fix}(J)) \geq \varepsilon \implies \|\psi_-(x)\| \geq m(\varepsilon).$$

Since ψ_- always vanishes on $\text{Fix}(J)$, qualitative separation only adds the implication $\psi_-(x) = 0 \Rightarrow Jx = x$. Quantitative separation says that points at distance at least ε from the fixed locus have odd readout at least $m(\varepsilon)$ in norm (with the convention $\text{dist}(x, \emptyset) = +\infty$). It implies that $\psi_-(x) = 0$ only if x lies in the closure of $\text{Fix}(J)$, so it implies qualitative separation when $\text{Fix}(J)$ is closed, for instance when J is continuous. The metric enters only through the function $x \mapsto \text{dist}(x, \text{Fix}(J))$; in Theorem 5.2 it may be replaced by any function $\delta : X \rightarrow [0, \infty]$ for which the displayed implication holds.

Lemma 4.3 (Trace identity). *Let $\hat{\mathfrak{X}}$ be an involutive object ledger with $\int_X \|\psi_-\|^2 d\mu < \infty$. Then the map $x \mapsto \psi_-(x) \psi_-(x)^*$ is Bochner integrable in $\mathcal{S}_1(Y)$, the operator A_X is positive and trace-class,*

$$\langle A_X h, k \rangle = \int_X \langle \psi_-(x), h \rangle \langle \psi_-(x), k \rangle d\mu(x) \quad (h, k \in Y),$$

and

$$\text{tr } A_X = \int_X \|\psi_-(x)\|^2 d\mu(x).$$

Proof. For $u, v \in Y$ we have $uu^* - vv^* = u(u - v)^* + (u - v)v^*$, and a rank-one operator ab^* has trace norm $\|a\| \|b\|$. Hence $\|uu^* - vv^*\|_1 \leq (\|u\| + \|v\|) \|u - v\|$, so $v \mapsto vv^*$ is continuous from Y to $\mathcal{S}_1(Y)$. Composed with the strongly measurable map ψ_- , it gives a strongly measurable map $x \mapsto \psi_-(x) \psi_-(x)^*$ whose norm $\|\psi_-(x) \psi_-(x)^*\|_1 = \|\psi_-(x)\|^2$ is integrable by (T1). The map is therefore Bochner integrable and $A_X \in \mathcal{S}_1(Y)$.

Continuous linear functionals commute with Bochner integrals. Applying this to the functionals of (T2) gives the displayed formula for $\langle A_X h, k \rangle$. Its right-hand side is symmetric in h and k , and for $h = k$ it equals $\int_X \langle \psi_-, h \rangle^2 d\mu \geq 0$; so A_X is positive. Applying the same principle to the trace, which is a continuous linear functional by (T1), and using $\text{tr}(vv^*) = \|v\|^2$, gives $\text{tr } A_X = \int_X \|\psi_-\|^2 d\mu$. $\square$

By (T3) and the trace identity, the following are equivalent: $A_X = 0$; $\text{tr } A_X = 0$; $\psi_-(x) = 0$ for μ -almost every x . The last equivalence holds because the integral of the nonnegative function $\|\psi_-\|^2$ vanishes exactly when the function vanishes almost everywhere.

Remark 4.4 (Analysis operator). Under the hypothesis of Lemma 4.3, define $V : Y \rightarrow L^2(X, \mu)$ by $(Vh)(x) = \langle \psi_-(x), h \rangle$. By the Cauchy–Schwarz inequality $\|Vh\|^2 \leq \|h\|^2 \int_X \|\psi_-\|^2 d\mu$, so V is

bounded, and by the lemma $\langle A_X h, h \rangle = \|Vh\|^2 = \langle V^* V h, h \rangle$. Two self-adjoint operators with the same quadratic form are equal, so

$$A_X = V^* V.$$

In particular V is a Hilbert–Schmidt operator with $\|V\|_{\text{HS}}^2 = \text{tr } V^* V = \text{tr } A_X = \int_X \|\psi_-\|^2 d\mu$. Thus A_X is the Gram operator of the odd readout, and domination questions about A_X are questions about V ; see Section 6.

5 Direct confinement

The trace identity turns information about A_X into information about where the mass of μ lies. The qualitative statement uses $A_X = 0$; the quantitative one uses only the size of $\text{tr } A_X$.

Theorem 5.1 (Separation–confinement). *Let $\hat{\mathfrak{X}}$ be an involutive object ledger with $\int_X \|\psi_-\|^2 d\mu < \infty$, and suppose $A_X = 0$. Then $\psi_-(x) = 0$ for μ -almost every x . If moreover ψ_- is qualitatively separating, then $Jx = x$ for μ -almost every x ; that is, $X \setminus \text{Fix}(J)$ is μ -null. In the finite weighted case, every object of positive weight is fixed by J .*

Proof. By Lemma 4.3, $\int_X \|\psi_-\|^2 d\mu = \text{tr } A_X = 0$. The integrand is nonnegative and measurable, so $\|\psi_-(x)\| = 0$ for μ -almost every x . If ψ_- is qualitatively separating, the set of x at which the separation equivalence fails is also μ -null; outside the union of the two null sets, $\psi_-(x) = 0$ and hence $Jx = x$. In the finite case the null sets contain no object of positive weight. $\square$

Theorem 5.2 (Quantitative confinement). *Let $\hat{\mathfrak{X}}$ be an involutive object ledger on a metric space X , with $\int_X \|\psi_-\|^2 d\mu < \infty$ and with ψ_- quantitatively separating with modulus m . Then, for every $\varepsilon > 0$,*

$$\mu^*(\{x \in X : \text{dist}(x, \text{Fix}(J)) \geq \varepsilon\}) \leq \frac{\text{tr } A_X}{m(\varepsilon)^2},$$

where μ^* is the outer measure of μ ; it equals μ on measurable sets. The same bound holds with $\text{dist}(\cdot, \text{Fix}(J))$ replaced by any function $\delta : X \rightarrow [0, \infty]$ such that $\delta(x) \geq \varepsilon$ implies $\|\psi_-(x)\| \geq m(\varepsilon)$.

Proof. Fix $\varepsilon > 0$ and put $E_\varepsilon = \{x : \text{dist}(x, \text{Fix}(J)) \geq \varepsilon\}$ and $F_\varepsilon = \{x : \|\psi_-(x)\| \geq m(\varepsilon)\}$. Quantitative separation gives $E_\varepsilon \subseteq F_\varepsilon$, and F_ε is measurable because ψ_- is strongly measurable (up to a μ -null set if ψ is only almost everywhere strongly measurable, which does not affect the bound). Markov’s inequality [4] and the trace identity give

$$m(\varepsilon)^2 \mu(F_\varepsilon) \leq \int_{F_\varepsilon} \|\psi_-\|^2 d\mu \leq \int_X \|\psi_-\|^2 d\mu = \text{tr } A_X.$$

Since $m(\varepsilon) > 0$, dividing and using $\mu^*(E_\varepsilon) \leq \mu(F_\varepsilon)$ gives the bound. The proof used the metric only through the inclusion $E_\varepsilon \subseteq F_\varepsilon$, which holds for any δ as in the statement. $\square$

Example (Complex conjugation, continued). In the example of Section 3, $P_-(u, v) = (0, v)$, so $\psi_-(z) = (0, \text{Im } z)$ and

$$A_X = \left(\int_X (\text{Im } z)^2 d\mu(z) \right) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

provided the integral is finite. If X is a disc centred on the real axis, then $\text{Re } z \in X$ for every $z \in X$, so $\text{dist}(z, X \cap \mathbb{R}) = |\text{Im } z|$; the readout is then quantitatively separating with $m(\varepsilon) = \varepsilon$, and Theorem 5.2 becomes the familiar Markov–Chebyshev bound $\mu\{|\text{Im } z| \geq \varepsilon\} \leq \varepsilon^{-2} \int_X (\text{Im } z)^2 d\mu$.

The confinement theorem of Section 7 supplies the missing input $A_X = 0$: it derives it from domination by operators of vanishing trace, after which Theorem 5.1 applies. The quantitative theorem is useful when only an upper bound on $\text{tr } A_X$ is available, as in Proposition 8.4.

6 Domination and Douglas factorization

Section 5 shows what follows once A_X vanishes. To force the vanishing, the confinement theorem uses upper bounds on A_X in the Loewner order. In applications such bounds typically come in two pieces: a main term, which an argument specific to the application makes small, and an error term, kept explicit.

Definition 6.1 (Completed domination bridge). Let $\hat{\mathfrak{X}}$ be an involutive object ledger with $\int_X \|\psi_-\|^2 d\mu < \infty$. A *completed domination bridge* for A_X is a pair of positive trace-class operators K^- and E on Y such that

$$A_X \preceq K^- + E.$$

The operator K^- is the *main term* and E the *defect*. The bridge is *exact* when $E = 0$. A sequence of bridges $A_X \preceq K_n^- + E_n$ is a *domination-record sequence* if the bounds $B_n = K_n^- + E_n$ satisfy $\text{tr } B_n \rightarrow 0$ as $n \rightarrow \infty$.

A bridge is bookkeeping rather than a theorem: it records which part of an upper bound is the controlled main term and which part is error. As a definition it imposes nothing beyond $A_X \preceq K^- + E$; indeed $K^- = A_X$, $E = 0$ is always an exact bridge. Its content in applications is that K^- is computed from data that do not involve ψ_- . The notation follows the framework's companion papers, where the main term is computed on the carrier side from other data and the bridge inequality is established by a separate argument. For example, in the finite-dimensional setting of [15], $K^- = L^- C^\dagger (L^-)^*$, where C is a positive audit operator on a carrier space, $C^\dagger$ its Moore–Penrose pseudoinverse, and L^- an anti-invariant probe with $\ker C \subseteq \ker L^-$; the minus sign records that K^- is the currency of the anti-invariant probe. A domination-record sequence is exactly the input of the confinement theorem (Theorem 7.1). Exact bridges have a useful reformulation, which rests on the following classical theorem of Douglas.

Theorem 6.2 (Douglas factorization [3]). *Let H , K_1 and K_2 be real Hilbert spaces and let $V : H \rightarrow K_1$ and $W : H \rightarrow K_2$ be bounded linear operators. Then*

$$V^*V \preceq W^*W \iff V = TW \text{ for some bounded } T : K_2 \rightarrow K_1 \text{ with } \|T\| \leq 1.$$

Proof. If $V = TW$ with $\|T\| \leq 1$, then for every $h \in H$,

$$\langle V^*Vh, h \rangle = \|TWh\|^2 \leq \|Wh\|^2 = \langle W^*Wh, h \rangle,$$

so $V^*V \preceq W^*W$. Conversely, suppose $V^*V \preceq W^*W$, so that $\|Vh\| \leq \|Wh\|$ for every $h \in H$. Define T_0 on the range of W by $T_0(Wh) = Vh$. It is well defined and linear: if $Wh = Wh'$ then $\|Vh - Vh'\| \leq \|W(h - h')\| = 0$. It is contractive by the same inequality. Hence T_0 extends by continuity to a contraction on the closure $\overline{\text{Ran } W}$. Setting $T = T_0$ on $\overline{\text{Ran } W}$ and $T = 0$ on its orthogonal complement gives a bounded operator $T : K_2 \rightarrow K_1$ with $\|T\| \leq 1$ and $TW = V$. $\square$

The theorem is due to Douglas [3], who proves, for bounded operators A, B on a single complex Hilbert space, that $AA^* \preceq \lambda^2 BB^*$ for some $\lambda \geq 0$ if and only if $A = BC$ for a bounded operator C , and that C can then be chosen with $\|C\| \leq \lambda$. With $\lambda = 1$, $A = V^*$ and $B = W^*$ this is the displayed statement after taking adjoints. The short proof above, which is standard, covers real scalars and operators between different spaces.

Exact bridges as factorizations. By Remark 4.4, $A_X = V^*V$ for the analysis operator $V : Y \rightarrow L^2(X, \mu)$, $(Vh)(x) = \langle \psi_-(x), h \rangle$. Every positive operator K^- on Y can be written as $K^- = W^*W$, for instance with $W = (K^-)^{1/2}$, which is Hilbert–Schmidt when K^- is trace-class. Theorem 6.2 therefore says that $A_X \preceq K^-$ exactly when the analysis operator of the odd readout factors as $V = TW$ through a contraction T . In words: an exact bridge exists precisely when every odd measurement $x \mapsto \langle \psi_-(x), h \rangle$ is obtained from the main-term measurement Wh by one fixed contraction.

7 The duality-confinement membrane theorem

This section proves the confinement theorem. Its input is a sequence of domination records whose traces vanish; its output is the almost-everywhere vanishing of the odd readout and, when the readout is separating, confinement of the mass of μ to the fixed locus. In the framework’s companion papers a positive budget of this kind, bounding a carrier-side quantity at every stage, is called a *membrane* [15]; the theorem’s name records that it is the step turning such budgets into confinement. In the vocabulary of Section 2, it is the mathematical consequence of the structural law Γ_{SDTC} : it does not produce the domination records, it uses them.

Theorem 7.1 (Duality-confinement membrane theorem). *Let $\hat{\mathfrak{X}} = (X, J, \mu, \psi, Y, J_{\text{iso}})$ be an involutive object ledger with $\int_X \|\psi_-\|^2 d\mu < \infty$. Suppose there are positive trace-class operators B_n on Y such that*

$$A_X \preceq B_n \quad (n = 1, 2, \dots) \quad \text{and} \quad \text{tr } B_n \rightarrow 0 \quad (n \rightarrow \infty).$$

Then $A_X = 0$ and $\psi_-(x) = 0$ for μ -almost every x . If, in addition, ψ_- is qualitatively separating, then $Jx = x$ for μ -almost every x , so the mass of μ is confined to $\text{Fix}(J)$; in the finite weighted case, every object of positive weight is fixed by J .

Conversely, if $\psi_-(x) = 0$ for μ -almost every x —in particular, if $Jx = x$ for μ -almost every x —then $A_X = 0$, and the constant sequence $B_n = 0$ satisfies the hypotheses.

Proof. By Lemma 4.3, A_X is positive and trace-class. For each n , (T4) applied to $0 \preceq A_X \preceq B_n$ gives

$$0 \leq \text{tr } A_X \leq \text{tr } B_n.$$

The middle quantity does not depend on n and the right-hand side tends to zero, so $\text{tr } A_X = 0$. By (T3), $A_X = 0$. The remaining direct assertions are Theorem 5.1. For the converse, ψ_- vanishes on $\text{Fix}(J)$, so almost-everywhere fixedness implies $\psi_- = 0$ almost everywhere; then $\text{tr } A_X = \int_X \|\psi_-\|^2 d\mu = 0$, so $A_X = 0$ by (T3), and $A_X \preceq 0$. $\square$

The converse shows that, for a given ledger, trace-vanishing domination records exist if and only if ψ_- vanishes almost everywhere, and hence, for a separating readout, if and only if the mass is confined to the fixed locus. The hypothesis is therefore equivalent to the conclusion, and the theorem does not make confinement easier to prove. Its role is to fix the form in which confinement is to be established when A_X is known only through a bridge $A_X \preceq K^- + E$ to operators computed from other data (Section 6): then trace bounds on K^- and E , rather than on A_X itself, are what must be proved.

Two features of the proof are worth isolating. First, it uses A_X only through four properties: A_X is positive; the trace is monotone for the Loewner order; the trace of a positive operator is nonnegative; and a positive operator of trace zero is zero. Second, the geometric conclusion enters only at the end, through the trace identity and separation. The first observation gives an abstract form of the theorem.

Remark 7.2 (Abstract form). Let C be a set with a distinguished element 0, a relation $\preceq$, a subset of *positive* elements, and a map $\tau : C \rightarrow \mathbb{R}$ such that $\tau(c) \geq 0$ for every positive c , and $a \preceq b$ implies $\tau(a) \leq \tau(b)$. Let $a \in C$ be positive with $\tau(a) = 0$ only if $a = 0$. If $a \preceq b_n$ for all n and $\tau(b_n) \rightarrow 0$, then $a = 0$: indeed $0 \leq \tau(a) \leq \tau(b_n) \rightarrow 0$. Theorem 7.1 is the case of positive trace-class operators with the Loewner order and the operator trace.

Where the theorem is used. A companion paper [16] considers the involution $s \mapsto 1 - \bar{s}$ on the zeros of the Riemann zeta function. It states, as an explicit hypothesis, a recognition source supplying domination records for an odd readout of the zeros, and applies the fixed-locus conclusion of the theorem to that ledger. The present paper uses nothing from that work.

8 Exhaustive squeeze and defected budgets

In applications the domination records are rarely available on Y itself. More often one controls a finite piece of the problem—a window, a truncation, a finite-dimensional subspace—and must account separately for what the piece misses. This section records the two refinements of the confinement theorem needed in that situation. Throughout, $\hat{\mathfrak{X}}$ is an involutive object ledger with $\int_X \|\psi_-\|^2 d\mu < \infty$.

Definition 8.1 (Exhaustive moving ledger). An *exhaustive moving ledger* for A_X consists of, for each $n \geq 1$, a real Hilbert space Y_n , a bounded operator $\iota_n : Y_n \rightarrow Y$, a positive trace-class operator $A_{X,n}$ on Y_n and a positive trace-class operator T_n on Y , such that

$$A_X \preceq \iota_n A_{X,n} \iota_n^* + T_n \quad (n \geq 1).$$

It is *vanishing-exhaustive* if there are positive trace-class operators B_n on Y_n with $A_{X,n} \preceq B_n$ and

$$\text{tr}(\iota_n B_n \iota_n^*) + \text{tr} T_n \rightarrow 0 \quad (n \rightarrow \infty).$$

The operator $A_{X,n}$ is the part of the ledger seen by the n -th window, ι_n transports it into Y , and T_n is a positive tail accounting for what the window misses. Despite the name, no exhaustion, monotonicity or convergence of the windows is required; the name reflects the typical use, illustrated next.

Theorem 8.2 (Exhaustive ledger squeeze). *If A_X admits a vanishing-exhaustive moving ledger, then $A_X = 0$ and $\psi_-(x) = 0$ for μ -almost every x . If, in addition, ψ_- is qualitatively separating, then $Jx = x$ for μ -almost every x .*

Proof. Conjugation preserves the Loewner order: if $A \preceq B$ on Y_n , then for $h \in Y$, $\langle \iota_n(B-A)\iota_n^* h, h \rangle = \langle (B-A)\iota_n^* h, \iota_n^* h \rangle \geq 0$, so $\iota_n A \iota_n^* \preceq \iota_n B \iota_n^*$. Hence

$$A_X \preceq \iota_n A_{X,n} \iota_n^* + T_n \preceq \iota_n B_n \iota_n^* + T_n =: B'_n.$$

Each B'_n is positive and trace-class, because the trace class is an ideal, and $\text{tr} B'_n \rightarrow 0$ by hypothesis. Theorem 7.1 applied to the sequence (B'_n) gives the conclusions. $\square$

Theorem 8.2 is Theorem 7.1 applied to the records B'_n ; its content is bookkeeping, recording the data an application typically computes.

Example (Finite-dimensional windows). Let Y be separable, and let P_n be orthogonal projections onto finite-dimensional subspaces $Y_n \subseteq Y$ with $P_n \rightarrow I$ strongly, let ι_n be the inclusion of Y_n , so that $\iota_n \iota_n^* = P_n$, and put $Q_n = I - P_n$. With $A_X = V^*V$ as in Remark 4.4, the case $t = 1$ of the operator inequality displayed below, applied to $V = VP_n + VQ_n$, gives

$$A_X \preceq 2P_n A_X P_n + 2Q_n A_X Q_n.$$

So the compressions $A_{X,n} = 2\iota_n^* A_X \iota_n$ and the tails $T_n = 2Q_n A_X Q_n$ form an exhaustive moving ledger. The tails satisfy $\text{tr } T_n = 2\|VQ_n\|_{\text{HS}}^2 \rightarrow 0$ by dominated convergence, because V is Hilbert-Schmidt. Whether the window bounds B_n can be chosen with $\text{tr}(\iota_n B_n \iota_n^*) \rightarrow 0$ is the substantive question in any application.

The second refinement concerns a bound with two error terms of different origin. Such bounds come from the operator form of the elementary inequality $(\alpha + \beta)^2 \leq (1+t)\alpha^2 + (1+t^{-1})\beta^2$: for bounded operators V_1, V_2 and $t > 0$,

$$(1+t)V_1^*V_1 + (1+t^{-1})V_2^*V_2 - (V_1 + V_2)^*(V_1 + V_2) = (\sqrt{t}V_1 - V_2/\sqrt{t})^*(\sqrt{t}V_1 - V_2/\sqrt{t}) \succeq 0.$$

So if the analysis operator of Remark 4.4 splits as $V = V_1 + V_2$, then A_X is dominated, for every $t > 0$, by $(1+t)V_1^*V_1 + (1+t^{-1})V_2^*V_2$.

Definition 8.3 (Defected obstruction budget). A *defected obstruction budget* for A_X consists of positive trace-class operators K^- , E_{src} and E_{br} on Y such that

$$A_X \preceq (1+t)(K^- + E_{\text{src}}) + (1+t^{-1})E_{\text{br}} \quad \text{for every } t > 0.$$

Here E_{src} is a source defect and E_{br} a bridge defect. A *sequence* of defected obstruction budgets is a sequence $(K_n^-, E_{\text{src},n}, E_{\text{br},n})$, each a defected obstruction budget for the same operator A_X .

The operators K^- and E_{src} enter only through their sum; they are kept apart because in applications they have different origins, a computed main term and a defect in its source.

The choice $t = 1$ gives the unweighted bound $2(K^- + E_{\text{src}}) + 2E_{\text{br}}$; the next proposition optimizes over t .

Proposition 8.4 (Optimized scalar trace budget). *Let $(K^-, E_{\text{src}}, E_{\text{br}})$ be a defected obstruction budget for A_X , and put $a = \text{tr}(K^- + E_{\text{src}})$ and $b = \text{tr } E_{\text{br}}$. Then $a, b \geq 0$,*

$$\inf_{t>0} [(1+t)a + (1+t^{-1})b] = (\sqrt{a} + \sqrt{b})^2,$$

the infimum being attained at $t = \sqrt{b/a}$ when $a, b > 0$, and

$$\text{tr } A_X \leq (\sqrt{a} + \sqrt{b})^2.$$

Consequently, if $(K_n^-, E_{\text{src},n}, E_{\text{br},n})$ is a sequence of defected obstruction budgets for A_X with $a_n = \text{tr}(K_n^- + E_{\text{src},n}) \rightarrow 0$ and $b_n = \text{tr } E_{\text{br},n} \rightarrow 0$, then all conclusions of Theorem 7.1 hold.

Proof. The traces a and b are nonnegative by (T3). For $t > 0$,

$$(1+t)a + (1+t^{-1})b - (\sqrt{a} + \sqrt{b})^2 = ta + \frac{b}{t} - 2\sqrt{ab} = \left(\sqrt{ta} - \sqrt{b/t}\right)^2 \geq 0,$$

with equality when $ta = b/t$, that is, at $t = \sqrt{b/a}$ if $a, b > 0$. If $a = 0$ the budget equals $b + b/t$, which decreases to $b = (\sqrt{a} + \sqrt{b})^2$ as $t \rightarrow \infty$; if $b = 0$ it equals $a + a/t$, which decreases to a as $t \rightarrow 0$. This proves the formula for the infimum. For each $t > 0$, (T4) and linearity of the trace give $\text{tr } A_X \leq (1+t)a + (1+t^{-1})b$; taking the infimum gives $\text{tr } A_X \leq (\sqrt{a} + \sqrt{b})^2$. For a sequence of budgets, $0 \leq \text{tr } A_X \leq (\sqrt{a_n} + \sqrt{b_n})^2 \rightarrow 0$, so $\text{tr } A_X = 0$, hence $A_X = 0$ by (T3), and the remaining conclusions follow from Theorem 5.1. $\square$

When $A_X = V^*V$ and $V = V_1 + V_2$ with $K^- + E_{\text{src}} = V_1^*V_1$ and $E_{\text{br}} = V_2^*V_2$, as in the operator inequality above, the bound $\text{tr } A_X \leq (\sqrt{a} + \sqrt{b})^2$ is the triangle inequality $\|V\|_{\text{HS}} \leq \|V_1\|_{\text{HS}} + \|V_2\|_{\text{HS}}$, and the optimization over t is a standard proof of it. The proposition applies more generally, to any family of bounds of the displayed form.

For a single budget, combining the proposition with Theorem 5.2 gives the explicit bound $\mu^*\{x : \text{dist}(x, \text{Fix}(J)) \geq \varepsilon\} \leq (\sqrt{a} + \sqrt{b})^2 / m(\varepsilon)^2$ whenever ψ_- is quantitatively separating with modulus m .

9 Scope and non-claims

This section states what the paper does not claim.

SDTC is a hypothesis. Self-Dual Trace Confinement is a named hypothesis about closures with genuine involutive self-duality (Section 2). It is not derived from anything in this paper, and it is not established in any particular instance. By the converse part of Theorem 7.1, in a given ledger the existence of trace-vanishing domination records is equivalent to the almost-everywhere vanishing of the odd readout, and hence, when the odd readout is separating, to confinement. So SDTC, in any instance with a separating readout, asserts as much as the confinement statement it is used to prove; its role is to name the form in which the confinement is to be established.

Infinite-dimensional response spaces. All results are proved in the text for an arbitrary real Hilbert space Y , using the standard trace-class facts (T1)–(T4) of Section 4. The formal verification also covers arbitrary dimension, with the trace of a positive operator taken as its diagonal sum in a Hilbert basis; the facts (T1)–(T4) themselves, Remark 4.4 and its use in Section 6 are not formally verified, and Appendix A lists the correspondence item by item.

Possible instances are open problems. The instances discussed in Section 10 indicate where the framework might apply. None is claimed as a result: in each case the domination records would have to be produced by an argument specific to the instance.

No novelty is claimed for classical operator theory. The Loewner order [7, 1, 6], trace ideals [9, 8], Markov’s inequality [4], the arithmetic–geometric mean inequality [5] and Douglas’s factorization theorem [3] are classical, and the proofs given here are standard. The contribution of the paper is the formulation: the involutive object ledger, the odd readout and its Gram operator A_X , the bookkeeping of domination records with explicit defects, and the reduction of fixed-locus confinement to trace estimates.

10 Discussion

A worked instance: self-adjointness. The definitions have a transparent instance in matrix analysis. Let X be the space $M_d(\mathbb{C})$ of complex $d \times d$ matrices, μ a finite positive Borel measure on X with $\int_X \|T\|_{\text{HS}}^2 d\mu(T) < \infty$, and $J(T) = T^*$. Regard $Y = M_d(\mathbb{C})$ as a real Hilbert space with the inner product $\langle A, B \rangle = \text{Re tr}(A^*B)$, let $J_{\text{iso}}(A) = A^*$, which is a real-linear isometric involution, and let ψ be the identity. Then

$$\psi_-(T) = \frac{1}{2}(T - T^*)$$

is the skew-Hermitian part of T , the fixed locus is the set of Hermitian matrices, and ψ_- is qualitatively separating. Since the Hermitian part $\frac{1}{2}(T + T^*)$ is the Hermitian matrix nearest to T in the Hilbert–Schmidt norm, $\text{dist}(T, \text{Fix}(J)) = \|\psi_-(T)\|$ and ψ_- is quantitatively separating with $m(\varepsilon) = \varepsilon$. Theorem 7.1 then says: μ is carried by the Hermitian matrices if and only if the Gram operator of the skew-Hermitian parts is dominated by positive operators of vanishing trace. Theorem 5.2 bounds the μ -mass of matrices at distance at least ε from the Hermitian ones by $\varepsilon^{-2} \text{tr } A_X$. Since $\text{tr } A_X = \int_X \|\psi_-\|^2 d\mu$ can be computed directly here, the instance illustrates the definitions but gains nothing from the domination formulation; a substantive use would need a situation in which A_X is known only through a bridge to independently computed operators.

Curves over finite fields. For a smooth projective curve over a finite field, the analogue of the Riemann hypothesis is a theorem of Weil [17, 18]. Stepanov introduced an elementary polynomial method and used it to bound the number of points on hyperelliptic curves over prime fields [10]; Bombieri, building on that method, gave an elementary proof of Weil’s theorem for all curves [2]. The zeta function of a curve of genus g over $\mathbb{F}_q$ has $2g$ inverse roots α , permuted by the involution $\alpha \mapsto q/\bar{\alpha}$ coming from the functional equation; its fixed locus is the circle $|\alpha| = \sqrt{q}$. The corresponding ledger is therefore a finite weighted one. Since the conclusion is known, trace-vanishing records exist trivially (the zero sequence), so the meaningful question is a different one: whether the Stepanov–Bombieri counting argument can be recast as a bridge $A_X \preceq K^- + E$ in which K^- and E are computed from point counts rather than from the roots. We have not carried this out.

Relation to classical operator theory. Stripped of the framework, Theorem 7.1 is the familiar fact that a positive trace-class operator dominated by operators of arbitrarily small trace is zero, applied to the Gram operator of an odd observable. What the formulation adds is a fixed interface: an application must specify the involution, the equivariant readout and the domination records, and the conclusion is then read off without further analysis.

Further directions. The natural direction is to find instances in which domination records can be produced by estimates, for instance through the factorization criterion of Section 6.

11 Conclusion

For an involution J of a measure space, a linear isometric involution of a real Hilbert space, and a strongly measurable equivariant readout with square-integrable odd part ψ_- , the Gram operator $A_X = \int_X \psi_- \psi_-^* d\mu$ is positive and trace-class with $\text{tr } A_X = \int_X \|\psi_-\|^2 d\mu$. Domination $A_X \preceq B_n$ by positive trace-class operators with $\text{tr } B_n \rightarrow 0$ forces $A_X = 0$, hence $\psi_- = 0$ almost everywhere, hence—when ψ_- is separating—confinement of the mass to the fixed locus $\text{Fix}(J)$; conversely, confinement makes the zero sequence a valid set of records. The same conclusion follows from domination on finite windows up to positive tails when the transported window bounds and the tails have traces tending to zero, and from two-defect budgets for A_X whose optimized trace $(\sqrt{a} + \sqrt{b})^2$ tends to zero. When the odd readout is quantitatively separating with a positive modulus m on a metric space, the outer measure of the points at distance at least ε from $\text{Fix}(J)$ is at most $\text{tr } A_X / m(\varepsilon)^2$. Douglas factorization identifies exact domination with contractive factorization of the analysis operator of ψ_- .

In the Six Birds framework these results are the mathematical side of the structural law Γ_{SDTC} , which asserts that closures with genuine involutive self-duality supply such domination records. The law is a hypothesis; the theorems of this paper say exactly what follows from it and what it amounts

to. The results, for real Hilbert spaces of arbitrary dimension, and the abstract squeeze are formally verified in Lean 4 with Mathlib (Appendix A).

A Formal verification

Every theorem, lemma and proposition of this paper, and the abstract squeeze of Remark 7.2, has been formally verified in the Lean 4 proof assistant (version 4.28.0) using the Mathlib library (revision 8f9d9cff6bd728b17a24e163c9402775d9e6a365), for real Hilbert spaces of arbitrary dimension. This appendix states how the objects of the paper are represented, what is checked only in the text, and which Lean theorem proves each numbered result.

Representation

The response space Y , the spaces H, K_1, K_2 of Theorem 6.2 and the window spaces Y_n of Theorem 8.2 are complete real inner product spaces; no finite dimension or separability is assumed. The operator J_{iso} is a linear isometric equivalence of Y that is an involution, the readout ψ is μ -almost everywhere strongly measurable, and the involution J need not be measurable. The ledger A_X is the Bochner integral of $x \mapsto \psi_-(x)\psi_-(x)^*$ in the operator norm. The trace of a positive operator T is the extended sum $\sum_i \langle Te_i, e_i \rangle \in [0, \infty]$ over a Hilbert basis (e_i) ; it is proved independent of the basis, and a positive trace-class operator is represented as a positive bounded operator whose trace is finite. The order $A \preceq B$ is the inequality $\langle Ah, h \rangle \leq \langle Bh, h \rangle$ of quadratic forms, which for the self-adjoint operators of the paper is the Loewner order. In Theorem 5.2 the distance to $\text{Fix}(J)$ is the extended distance, which is $+\infty$ when $\text{Fix}(J)$ is empty.

The formal proofs do not use the trace-class facts (T1)–(T4) of Section 4; they work directly with the operator norm Bochner integral and the extended trace.

What is checked only in the text

The following are not formally verified: that $x \mapsto \psi_-(x)\psi_-(x)^*$ is Bochner integrable in the trace class $\mathcal{S}_1(Y)$, and the facts (T1)–(T4) themselves, including the identification, for positive operators, of finite extended trace with membership in the trace class and of the extended sum with the trace (standard facts recalled in Section 4); the comparison of quantitative and qualitative separation after Definition 4.2; Remark 4.4 and the discussion of exact bridges as factorizations in Section 6; the Hilbert–Schmidt reading of Proposition 8.4; the worked examples; and the framework material of Section 2. The structural law Γ_{SDTC} is a hypothesis and is not formalized as a theorem.

Correspondence

Table 1. Numbered results and the Lean theorems that prove them for real Hilbert spaces of arbitrary dimension; names are relative to the namespace `SixBirdsDualityConfinement.DualityConfinement`. Definitions 4.2, 6.1, 8.1 and 8.3 record hypotheses; in Lean they appear as the hypotheses of the theorems listed for the results that use them. The modules `ConcreteFinite` and `ConcreteMeasured` contain separate proofs of the finite-dimensional cases with the linear-algebra trace.

Result	Lean theorems
Definitions 3.1 and 4.1	<code>ConcreteCore.pminus</code> , <code>ConcreteCore.psiMinus</code> , <code>ConcreteHilbert.ledger</code> , <code>ConcreteHilbert.PositiveTraceClass</code> , <code>ConcreteHilbert.positiveTraceClass_basis_iff</code> , <code>ConcreteHilbert.psiMinus_aestronglyMeasurable</code> , <code>ConcreteCore.psiMinus_equivariant</code> , <code>ConcreteCore.psiMinus_fixed</code>
Lemma 4.3	<code>ConcreteHilbert.ledger_positive</code> , <code>ConcreteHilbert.ledger_inner</code> , <code>ConcreteHilbert.ledger_trace</code> , <code>ConcreteHilbert.ledger_traceClass</code>
Theorem 5.1	<code>ConcreteHilbert.ledger_zero_iff</code> , <code>ConcreteHilbert.separation_confinement</code> , <code>ConcreteHilbert.atom_readout_of_zero</code> , <code>ConcreteHilbert.atom_fixed_of_zero</code>
Theorem 5.2	<code>ConcreteHilbert.quantitative_threshold</code> (arbitrary subsets, outer measure), <code>ConcreteHilbert.quantitative_distance</code>
Theorem 6.2	<code>ConcreteHilbert.douglas_factorization</code>
Theorem 7.1	<code>ConcreteHilbert.ledger_master</code> , <code>ConcreteHilbert.ledger_master_fixed</code> ; converse <code>ConcreteHilbert.ledger_records_iff</code> , <code>ConcreteHilbert.ledger_zero_of_ae</code> , <code>ConcreteHilbert.ledger_zero_of_ae_fixed</code> , <code>ConcreteHilbert.zero_budget_records</code>
Remark 7.2	<code>AbstractSqueeze.abstract_trace_squeeze</code>
Theorem 8.2	<code>ConcreteHilbert.ledger_exhaustive</code> , <code>ConcreteHilbert.ledger_exhaustive_fixed</code>
Proposition 8.4	<code>ConcreteHilbert.scalar_budget_infimum</code> , <code>ConcreteHilbert.optimized_budget_infimum</code> , <code>ConcreteHilbert.ledger_optimized_budget</code> , <code>ConcreteHilbert.ledger_budget_sequence</code> , <code>ConcreteHilbert.ledger_budget_sequence_fixed</code> ; attainment at $t = \sqrt{b/a}$ from <code>youngTrace_optimum</code> in <code>SixBirdsNeedles.MainCore</code>

The general Hilbert-space theorems are derived from the operator ledger, positive trace, confinement, exhaustion, trace budget and Douglas factorization results of the companion library `SixBirdsNeedles.MainCore` [15].

Trust base and availability

Every theorem listed above depends only on the three standard axioms of Lean’s type theory with classical logic (`propext`, `Classical.choice`, `Quot.sound`); no `sorry` and no additional axioms occur. The development is available at <https://github.com/ioannist/six-birds-duality-confinement>. The same repository contains an earlier formulation of these results over an abstract “typed positive cone”, in which the trace identity, Douglas factorization and the arithmetic–geometric mean step are supplied as hypotheses; the companion Riemann-hypothesis development builds on that formulation. The claims of the present paper rest only on the theorems listed in Table 1.

B Version history

Every numbered result of Version 1 keeps its number and place in later versions, and its name except where noted below.

Version 3 (October 1, 2026)

- The results are now stated and proved for positive trace-class operators on an arbitrary real Hilbert space. Version 2 stated them for an abstract “typed positive cone” and described the trace-class case as an open extension. The trace identity (Lemma 4.3) is now proved from Bochner integration in the trace class; Version 2 took it as part of the data.
- The readout in Definition 3.1 is required to be strongly measurable; together with $\int_X \|\psi_-\|^2 d\mu < \infty$ this makes the Bochner integral defining A_X exist. Definitions 4.1, 6.1, 8.1 and 8.3 now specify the operators involved (positive trace-class operators, bounded transports between Hilbert spaces).
- The confinement conclusions are stated as “ $Jx = x$ for μ -almost every x ”, which needs no measurability of $\text{Fix}(J)$; Version 2 assumed $X \setminus \text{Fix}(J)$ measurable. Theorem 5.2 is stated for outer measure, so the far set need not be measurable, and it holds for any displacement function δ in place of the distance, provided $\delta(x) \geq \varepsilon$ implies $\|\psi_-(x)\| \geq m(\varepsilon) > 0$.
- Theorem 7.1 now includes its converse: trace-vanishing domination records exist if and only if the odd readout vanishes almost everywhere. The abstract, Sections 1, 2, 7 and 9 now state the consequences: the theorem does not make confinement easier to prove, its role is to fix the form of a bridge argument, and SDTC cannot hold for all involutive ledgers (the complex-conjugation example is a counterexample). Version 2 did not state the converse or these consequences.
- Theorem 6.2 is stated for bounded operators between real Hilbert spaces, with a complete proof and a precise account of Douglas’s original formulation; its name changed from “Douglas factorization, typed-cone form” to “Douglas factorization”. The new Remark 4.4 identifies A_X with V^*V for the analysis operator V of the odd readout.
- Proposition 8.4 is proved over the real numbers, including the cases $a = 0$ or $b = 0$, and now includes the consequence for sequences of budgets. Version 2 treated the arithmetic–geometric mean step as a hypothesis on an abstract scalar type.
- The formal verification was extended: every theorem, lemma and proposition is now proved in Lean 4 with Mathlib for real Hilbert spaces of arbitrary dimension, as is the abstract squeeze (Remark 7.2). Appendix A was rewritten accordingly.
- The exposition was rewritten for readers outside the framework: the introduction states the results with their hypotheses, Section 2 explains the framework terms, and worked examples were added (complex conjugation, self-adjointness of matrices). The Version 2 reference to a “V-Differential” classification, which is not defined in the cited literature, was removed, as was a paragraph on charge conjugation, CPT and gauge symmetries that made no claim. The discussion of curves over finite fields now states that the Riemann hypothesis there is a theorem of Weil, attributes the elementary method to Stepanov and its general form to Bombieri, and presents that setting as a test case rather than a prediction.

- Cross-references now name the kind of result they point to; in Version 2 references to definitions, lemmas, propositions and remarks were printed as “Theorem”. The DOI of Stepanov [10] in the bibliography was corrected.

Version 2 (September 28, 2026)

- The hypotheses of the measure-theoretic conclusions were made explicit: integrability of $\|\psi_-\|^2$, the trace identity, and the passage from almost-everywhere vanishing to a measure statement.
- The defected obstruction budget was restated as a family of domination records for the same operator A_X , one for each $t > 0$, which is what the trace bound of Proposition 8.4 requires.
- The description of the formal verification was corrected to distinguish the results derived in Lean from those whose inputs are supplied as hypotheses, and a numerical form of the trace squeeze was added.
- The description of the companion Riemann-hypothesis paper was corrected to state which of its hypotheses are supplied.

References

- [1] Rajendra Bhatia. *Matrix Analysis*, volume 169 of *Graduate Texts in Mathematics*. Springer, New York, 1997. ISBN 978-0-387-94846-1. doi:10.1007/978-1-4612-0653-8.
- [2] Enrico Bombieri. Counting points on curves over finite fields (d’après S. A. Stepanov). In *Séminaire Bourbaki, 25e année (1972/1973), Exp. no. 430*, volume 383 of *Lecture Notes in Mathematics*, pages 234–241. Springer-Verlag, Berlin, 1974.
- [3] R. G. Douglas. On Majorization, Factorization, and Range Inclusion of Operators on Hilbert Space. *Proceedings of the American Mathematical Society*, 17(2):413–415, 1966. doi:10.1090/S0002-9939-1966-0203464-1.
- [4] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. Wiley, New York, second edition, 1999. ISBN 978-0-471-31716-6.
- [5] G. H. Hardy, J. E. Littlewood, and G. Pólya. *Inequalities*. Cambridge University Press, Cambridge, second edition, 1952. ISBN 978-0-521-05206-1.
- [6] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, second edition, 2013. ISBN 978-0-521-54823-6.
- [7] Karl Löwner. Über monotone Matrixfunktionen. *Mathematische Zeitschrift*, 38(1):177–216, 1934. doi:10.1007/BF01170633.
- [8] Michael Reed and Barry Simon. *Methods of Modern Mathematical Physics I: Functional Analysis*. Academic Press, New York, revised and enlarged edition, 1980. ISBN 978-0-12-585050-6.
- [9] Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, second edition, 2005. ISBN 978-0-8218-3581-4. doi:10.1090/surv/120.
- [10] S. A. Stepanov. On the number of points of a hyperelliptic curve over a finite prime field. *Mathematics of the USSR-Izvestiya*, 3(5):1103–1114, 1969. doi:10.1070/IM1969v003n05ABEH000834.
- [11] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Oxford University Press, Oxford, second edition, 1986. ISBN 978-0-19-853369-6. Revised by D. R. Heath-Brown.
- [12] Ioannis Tsiokos. Six Birds: Foundations of Emergence Calculus. arXiv:2602.00134, 2026.
- [13] Ioannis Tsiokos. Six Birds Foundations II: Admissibility, Meta-Theory, and the Exact-Six Program. doi:10.5281/zenodo.19672278, 2026.
- [14] Ioannis Tsiokos. Six Birds Foundations III: A Finite Audited Interaction Calculus for SBT. doi:10.5281/zenodo.20124595, 2026.
- [15] Ioannis Tsiokos. Emergence IS the needle killer. doi:10.5281/zenodo.23086988, 2026. Version 4, October 1, 2026.
- [16] Ioannis Tsiokos. Riemann Hypothesis via self-dual trace confinement: A conditional closure. doi:10.5281/zenodo.23086975, 2026. Version 3, October 1, 2026.

- [17] André Weil. *Sur les courbes algébriques et les variétés qui s'en déduisent*, volume 1041 of *Actualités scientifiques et industrielles*. Hermann, Paris, 1948. Publications de l'Institut de mathématique de l'Université de Strasbourg, no. 7.
- [18] André Weil. *Variétés abéliennes et courbes algébriques*, volume 1064 of *Actualités scientifiques et industrielles*. Hermann, Paris, 1948. Publications de l'Institut de mathématique de l'Université de Strasbourg, no. 8.